

Chapter 6

Inverse Laplace Transforms

So far, we've taken differential equations and converted them into the Laplace Domain equivalent so we can

1. Form block diagrams
2. Perform basic analysis
 - (a) Initial Value
 - (b) Final Value
 - (c) Find the zeros and poles
3. Transform in the open-loop and closed-loop models

The transformation to the Laplace Domain can be reversed using a LUT with only a little effort in utilizing partial-fraction expansion, PFE. To directly utilize this approach, the open-loop transfer function must be strictly proper (the order of the numerator must be less than the order of the denominator). If the system is not strictly proper, perform long division once to form a mixed expression with a constant and a strictly proper remainder.

We represent the transfer function in open-loop form with the denominator represented in binomial form with real roots. Any complex roots are written as second order expressions.

For systems without complex poles

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{s+p_1} + \frac{k_n}{s+p_2} + \dots + \frac{k_n}{s+p_n}$$

For systems with real and complex poles

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{s+p_1} + \frac{k_2}{s+p_2} + \dots + \frac{k_n}{s+p_n} + \frac{k_ms + k_{m+1}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$$

And of course we may have repeated poles — both real and complex. For a system such as that the generic equation is

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{(s+p_1)^2} + \frac{k_2}{s+p_1} + \frac{k_3}{s+p_2} + \dots + \frac{k_n}{s+p_n} + \frac{k_ms + k_{m+1}}{(\alpha_2 s^2 + \alpha_1 s + \alpha_0)^2} + \frac{k_{m+2}s + k_{m+3}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$$

PFE can be difficult for more complex systems. Additionally, the solution begins to be less accurate with more poles. Systems with many repeated poles or even poles close to one another being to degrade the solution. Mathematically, those are **ill-conditioned** systems. Most physical systems may be accurately solved with PFE.

Let's take a look at some examples.

Example — Distinct Real Poles and an impulse input

Given:

$$\frac{Y(s)}{R(s)} = \frac{3}{(s+6)(s+2)}$$

Find: the $y(t)$ when $r(t) = \delta(t)$

Represent the output as a sum of partials

$$Y(s) = \frac{3}{(s+6)(s+2)} = \frac{k_1}{s+6} + \frac{k_2}{s+2}$$

To solve for the residue k_1 , multiply both sides by the divisor $s+6$ and then solve the equation for $s = -6$

$$\frac{3(s+6)}{(s+6)(s+2)} = \frac{k_1(s+6)}{s+6} + \frac{k_2(s+6)}{s+2}, s = -6$$

$$\frac{3}{(s+2)} = k_1 + \frac{k_2(s+6)}{s+2}, s = -6$$

$$\frac{3}{(-4)} = k_1 + \frac{k_2(0)}{-4}$$

$$k_1 = \frac{-3}{4}$$

Now, repeat the process for k_2

$$\frac{3(s+2)}{(s+6)(s+2)} = \frac{k_1(s+2)}{s+6} + \frac{k_2(s+2)}{s+2}, s = -2$$

$$\frac{3}{(s+6)} = \frac{k_1(s+2)}{s+6} + k_2, s = -2$$

$$\frac{3}{4} = \frac{k_1(0)}{4} + k_2$$

$$k_2 = \frac{3}{4}$$

So

$$Y(s) = \frac{\frac{-3}{4}}{s+6} + \frac{\frac{3}{4}}{s+2}$$

Now, use a LUT on the expressions to find the final solution

$$y(t) = \frac{-3}{4}e^{-6t} + \frac{3}{4}e^{-2t}$$

In this form, $y(t)$ may be evaluated for any value $0 \leq t < \infty$.

Example — Bipropor with Distinct Real Poles and an impulse input

Given:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 13s + 6}{s^2 + 9s + 8}$$

Find: $y(t)$ when $r(t) = \delta$

After long division

$$Y(s) = 1 + \frac{4s - 2}{s^2 + 9s + 8}$$

Then, use PFE on the denominator polynomial term

$$\frac{4s - 2}{s^2 + 9s + 8} = \frac{4s - 2}{(s + 8)(s + 1)} = \frac{k_1}{s + 8} + \frac{k_2}{s + 1}$$

So

$$k_1 = \frac{4s - 2}{s + 1}, s = -8$$

substitute $s = -8$ and simplify

$$k_1 = \frac{4(-8) - 2}{(-8) + 1} = \frac{-34}{-7} = \frac{34}{7}$$

and

$$k_2 = \frac{4s - 2}{s + 8}, s = -1$$

substitute $s = -1$ and simplify

$$k_2 = \frac{4(-1) - 2}{-1 + 8} = \frac{-6}{7}$$

Now

$$Y(s) = 1 + \frac{\frac{34}{7}}{s + 8} + \frac{\frac{-6}{7}}{s + 1}$$

And using a LUT

$$y(t) = \delta(t) + \frac{34}{7}e^{-8t} - \frac{6}{7}e^{-t}$$

Example — Repeated Real Poles and a Setpoint Input

Given:

$$\frac{Y(s)}{R(s)} = \frac{s + 72}{s^3 + 14s^2 + 57s + 72}$$

Find: $y(t)$ when $r(t) = 1$

Write $Y(s)$ as a sum of partials

$$Y(s) = \frac{s + 72}{s(s + 3)(s + 3)(s + 8)} = \frac{k_1}{s} + \frac{k_2}{(s + 3)^2} + \frac{k_3}{s + 3} + \frac{k_4}{s + 8}$$

Note the repeated residues k_2 and k_3 .

Residues k_1 , k_2 , and k_4 are solved as before. The results: $k_1 = 1$, $k_2 = -4.6$, and $k_4 = -0.32$. The only residue that cannot be solved as before is k_3 ; to solve it

$$k_3 = \left\{ \frac{d}{ds} \left[(s+3)^2 \frac{s+72}{s(s+3)^2(s+8)} \right] \right\}, s = -3$$

which simplifies to

$$k_3 = \left\{ \left[\frac{d}{ds} \frac{s+72}{s(s+8)} \right] \right\}, s = -3$$

Use the Quotient Rule to take the derivative

$$k_3 = \frac{(1)(s^2+8s) - (s+72)(2s+8)}{(s^2+8s)^2}, s = -3$$

$$k_3 = \frac{((-3)^2 - 24) - (72 - 3)(2(-3) + 8)}{((-3)^2 - 24)^2}$$

$$k_3 = \frac{(9 - 24) - (72 - 3)(8 - 6)}{(9 - 24)^2}$$

$$k_3 = \frac{-15 - (2 \cdot 69)}{225}$$

$$k_3 = \frac{-15 - 138}{225}$$

$$k_3 = \frac{-153}{225} = -0.68$$

Now that all the residues are determined

$$Y(s) = \frac{1}{s} - \frac{4.6}{(s+3)^2} - \frac{0.68}{s+3} - \frac{0.32}{s+8}$$

Take the inverse Laplace transform for the final solution

$$y(t) = 1 - 4.6te^{-3t} - 0.68e^{-3t} - 0.32e^{-8t}$$

Quotient Rule

When

$$f(x) = \frac{g(x)}{h(x)}$$

then

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{(h(x))^2}$$

6.1 MATLAB

The `residue` function in MATLAB is used to recompose a function into its partials.

For the previous example

$$\frac{Y(s)}{R(s)} = \frac{s+72}{s^3+14s^2+57s+72}$$

we define $B = [1, 72]$ and $A = [1, 14, 57, 72, 0]$ (note: setpoint input is distributed in A) then the function call

```
[r, p, k] = residue(B,A)
```

produces the residues r , poles p , and gain k that match the calculated values from the example

```
>> [r,p,k] = residue(B,A)
```

```
r =
```

```
-0.3200  
-0.6800  
-4.6000  
1.0000
```

```
p =
```

```
-8.0000  
-3.0000  
-3.0000  
0
```

```
k =
```

```
[]
```